

Worksheet -1



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Module Title: Programming in C++

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Task 1 : Programming Exercises:[Data types and Conditional Statements]

1. Write a program that takes a temperature value from the user. It should then allow the user to choose between Celsius (C) and Fahrenheit (F) for conversion. After the user selection, it should then convert the entered temperature to the chosen scale and display the result.

Use appropriate data types for temperature and handle error like non-numeric input.

Use the following formula for conversion:

$$F = (C \times 9/5) + 32$$

$$C = (F - 32) \times 5/9$$

➤ **Answer:**

```
#include <iostream>
using namespace std;
int main()
{
    int a;
    float temp,cal;
    cout<<"press accordingly for conversion"<<endl<<"1.Press 1 to convert into
    celcius"<<endl<<"2.Press 2 to convert into farhenheit"<<endl;
    cin>>a;
    if(a==1)
    {
        cout<<"Enter the data in farhenheit"<<endl;
        cin>>temp;
        cal=(temp-32)*5/9;
        cout<<"The temperature in celcius is:"<<cal<<"C"<<endl;

    }
    else if(a==2)
    {
        cout<<"Enter the data in celcius"<<endl;
        cin>>temp;
        cal=(temp*9/5)+32;
        cout<<"The temperature in farhenheit is:"<<cal<<"F"<<endl;
    }
    else
    {
        cout<<"Invalid input!"<<endl;
    }
    return 0;
}
```

❖ Output

```
"C:\Users\beena\OneDrive\De  X + v
press accordingly for conversion
1.Press 1 to convert into celcius
2.Press 2 to convert into farhenheit
1
Enter the data in farhenheit
78
The temperature in celcius is:25.5556C

Process returned 0 (0x0)   execution time : 7.365 s
Press any key to continue.
|
```

```
"C:\Users\beena\OneDrive\De  X + v
press accordingly for conversion
1.Press 1 to convert into celcius
2.Press 2 to convert into farhenheit
2
Enter the data in celcius
8
The temperature in farhenheit is:46.4F

Process returned 0 (0x0)   execution time : 6.956 s
Press any key to continue.
|
```

2. Write a C++ program to implement a number guessing game with different difficulty levels.

Easy difficulty ranges from 1-8, medium from 1-30, hard from 1-50. Then, generate a random number to check if the guess is correct based on the user's selection.

➤ **Answer:**

```
#include <iostream>
#include<cstdlib>
#include<ctime>
using namespace std;
int main()
{
    int level;
    int guess;
    int max;
    int number;

    cout<<"Welcome to number guessing game"<<endl;
    cout<<"Choose your level of difficulty"<<endl;
    cout<<"1.Easy(1-8)"<<endl;
    cout<<"2.Medium (1-30)"<<endl;
    cout<<"3.Hard (1-50)"<<endl;
    cout<<"Press to choose!"<<endl;
    cin>>level;

    switch(level)
    {
        case 1:
            max=8;
            break;
```

case 2:

max=30;

break;

case 3:

max=50;

break;

default:

cout<<"Invalid entry"<<endl;

break;

}

srand(time(0));

number=(rand()%max)+1;

do {

cout << "Guess a number between 1 and " << max << ": ";

cin >> guess;

if (guess < 1 || guess > max)

{

cout << "Out of range! Try again." << endl;

}

else if (guess < number)

{

cout << "Think higher!" << endl;

}

```

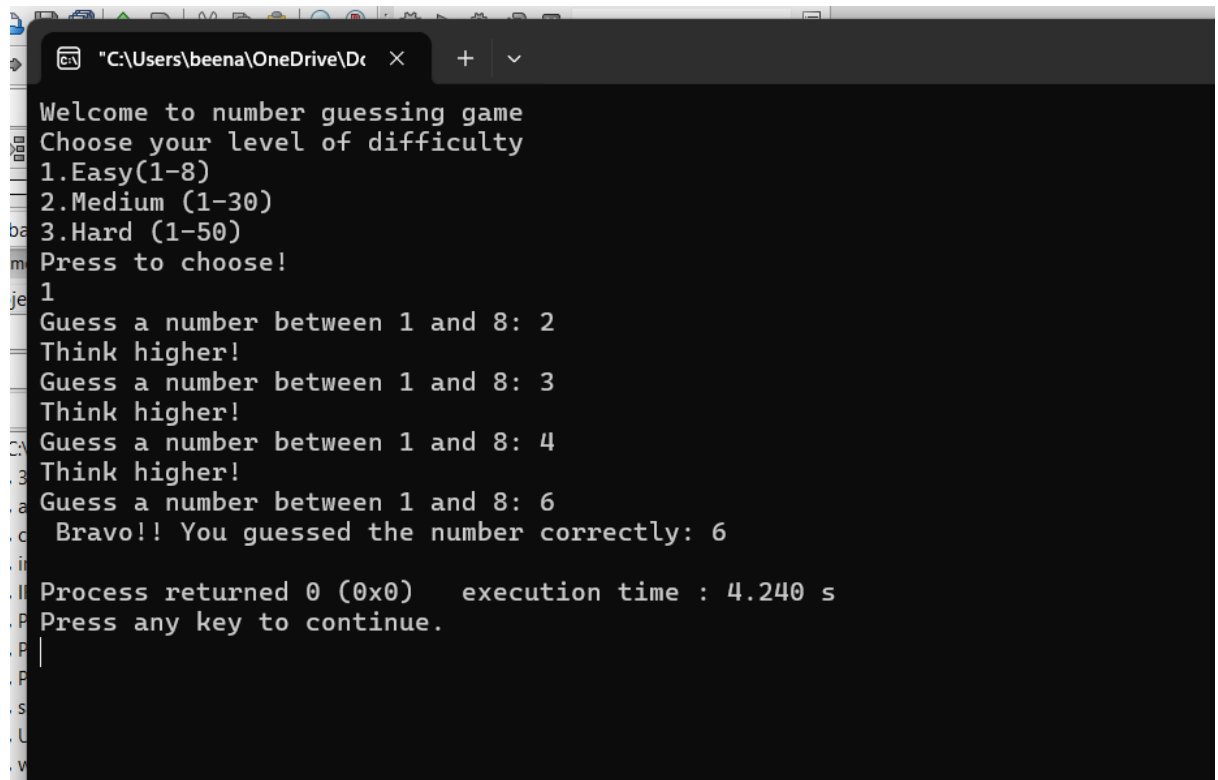
else if (guess > number)
{
    cout << "Stay low, stay grounded!" << endl;
}
else
{
    cout << " Bravo!! You guessed the number correctly: " << number << endl;
}

} while (guess != number);

return 0;
}

```

❖ Output



```

Welcome to number guessing game
Choose your level of difficulty
1.Easy(1-8)
2.Medium (1-30)
3.Hard (1-50)
Press to choose!
1
Guess a number between 1 and 8: 2
Think higher!
Guess a number between 1 and 8: 3
Think higher!
Guess a number between 1 and 8: 4
Think higher!
Guess a number between 1 and 8: 6
Bravo!! You guessed the number correctly: 6

Process returned 0 (0x0) execution time : 4.240 s
Press any key to continue.

```

```
"C:\Users\beena\OneDrive\De... x + v
Welcome to number guessing game
Choose your level of difficulty
1.Easy(1-8)
2.Medium (1-30)
3.Hard (1-50)
Press to choose!
2
Guess a number between 1 and 30
33
out of range

Process returned 0 (0x0)   execution time : 4.951 s
Press any key to continue.
|
```

3. Write a program that reads an array of integer numbers from the user and sorts the numbers in the ascending order.

➤ Answer:

```
#include<iostream>
using namespace std;
int main()
{
    int x;
    int a[100],temp=0;
    cout<<"how many numbers do you want to enter?"<<endl;
    cin>>x;
    cout<<"Enter the number"<<endl;
    for(int i=0;i<x;i++)
    {
        cin>>a[i];
    }
    for (int i = 0; i < x - 1; i++)
    {
        for (int j = i + 1; j < x; j++)
        {
```

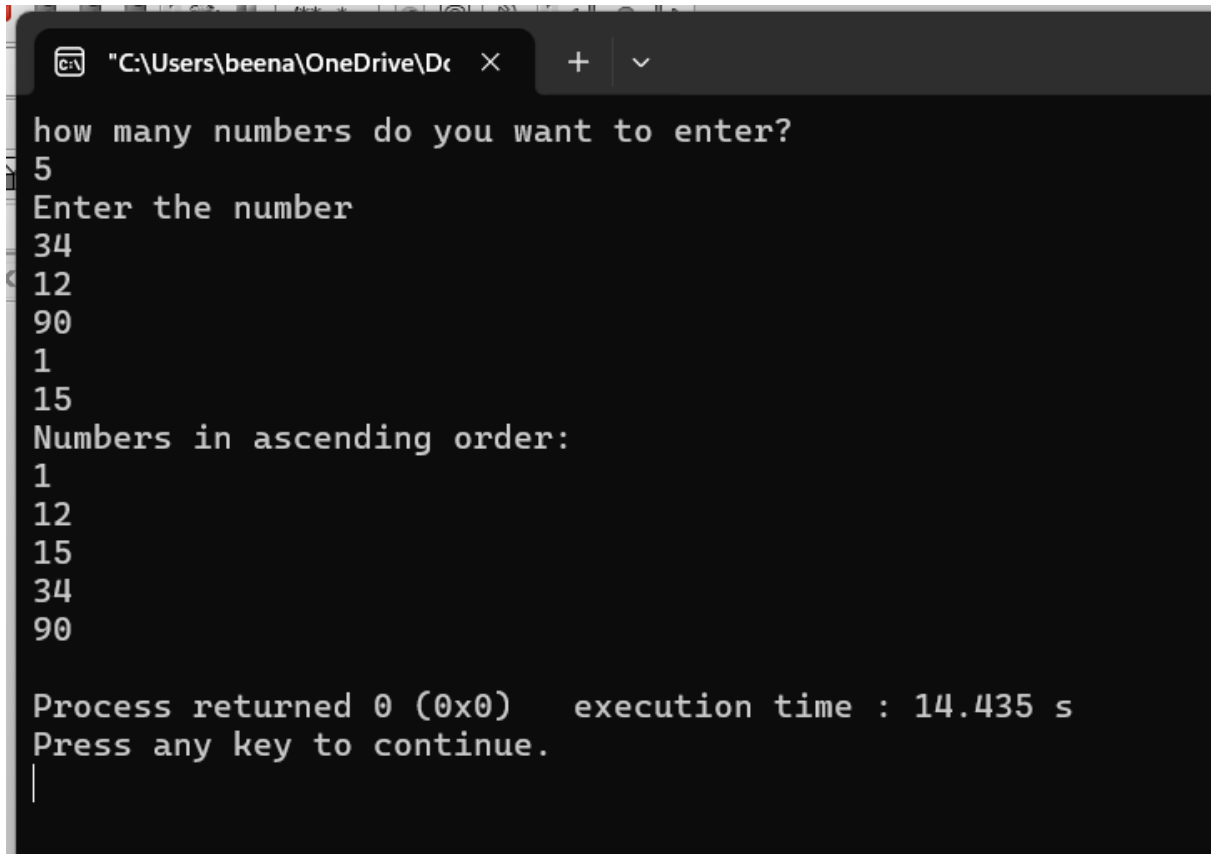
```

        if (a[i] > a[j])
        {
            temp = a[i];
            a[i] = a[j];
            a[j] = temp;
        }
    }
}
cout << "Numbers in ascending order:"<<endl;
for (int i = 0; i < x; i++)
{
    cout << a[i] << "\n" ;
}

return 0;
}

```

❖ Output



```

C:\Users\beena\OneDrive\De...
how many numbers do you want to enter?
5
Enter the number
34
12
90
1
15
Numbers in ascending order:
1
12
15
34
90

Process returned 0 (0x0)    execution time : 14.435 s
Press any key to continue.
|

```

4. Write a program that reads a number from the user and based on the user input, it says what day of the week it is, Sundays being 1 and Saturdays being 7. Your system should give appropriate response for invalid input entries.

➤ Answer:

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
int a;
```

```
    cout << "Press the number 1-7 to know what day of week it is" << endl;
```

```
    cin>>a;
```

```
    switch(a)
```

```
    {
```

```
        case 1:
```

```
        cout<<"Today is sunday"<<endl;
```

```
        break;
```

```
        case 2:
```

```
        cout<<"Today is Monday"<<endl;
```

```
        break;
```

```
        case 3:
```

```
        cout<<"Today is Tuesday"<<endl;
```

```
        break;
```

```
        case 4:
```

```
        cout<<"Today is Wednesday"<<endl;
```

```
        break;
```

```
        case 5:
```

```
        cout<<"Today is Thursday"<<endl;
```

```
        break;
```

```
        case 6:
```

```
        cout<<"Thankgod its Friday"<<endl;
```

```
        break;
```

```
        case 7:
```

```
        cout<<"Today is Saturday"<<endl;
```

```
        break;
```

```
        default:
```

```
        cout<<"There are only 7 days so number entered should not exceed 7
```

```
"<<endl;
```

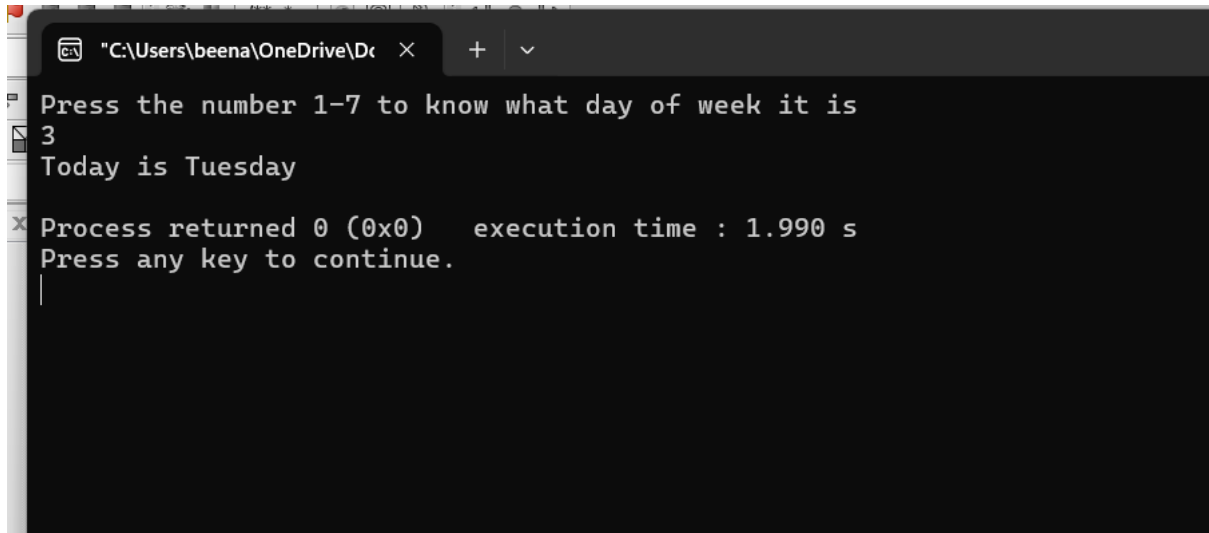
```
        break;
```

```

    }
    return 0;
}

```

❖ Output



```

"C:\Users\beena\OneDrive\De... X + v
Press the number 1-7 to know what day of week it is
3
Today is Tuesday
Process returned 0 (0x0) execution time : 1.990 s
Press any key to continue.

```

Task 2: Programming Exercises:[Control Statements]

1. Create a program that takes a positive integer as input and determines whether it's a "bouncy number". A bouncy number is one where the digits neither consistently increase nor consistently decrease when read from left to right. For example:

- 123 is NOT bouncy (digits consistently increase)
- 321 is NOT bouncy (digits consistently decrease)
- 120 is bouncy (neither consistently increasing nor decreasing)

➤ Answer:

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
int number;
```

```
int D1;
```

```
int D2;
```

```
cout<<"Enter a number"<<endl;
```

```
cin>>number;
```

```
if(number<100)
```

```
{
```

```

        cout<<"out of range! Bouncy numbers has atleast 3 digit"<<endl;

    }
    bool increasing=false;
    bool decreasing=false;

    D1= number % 10;
    number = number / 10;

    while (number > 0)
    {
        D2 = number % 10;

        if (D2 < D1)
        {
            increasing = true;
        }
        else if (D2 > D1)
        {
            decreasing = true;
        }

        D1 = D2;
        number = number / 10;

        if (increasing && decreasing)
        {
            break;
        }

    }

    if (increasing && decreasing)
    {
        cout << "It is a BOUNCY number." << endl;

    }
    else
    {
        cout << "It is not a Bouncy number." << endl;
    }
}

```

```
return 0;  
}
```

❖ Output

```
"C:\Users\beena\OneDrive\Dr...  
Enter a number  
67483  
It is a BOUNCY number.  
Process returned 0 (0x0) execution time : 6.019 s  
Press any key to continue.  
|
```

```
"C:\Users\beena\OneDrive\Dr...  
Enter a number  
1234  
It is not a Bouncy number.  
Process returned 0 (0x0) execution time : 4.006 s  
Press any key to continue.  
|
```

Task 3: Programming Exercises on Arrays

1. Write a program that manages a cinema ticket booking system. The program should display a 5x5 seating arrangement where:

1. Available seats are marked with 'O'
2. Booked seats are marked with 'X'

Program should:

1. Display the current seating arrangement
2. Ask user for row and column number (1-5) for booking
3. Mark that seat as booked ('X')
4. Show updated seating after each booking
5. Display error if user selects already booked seat
6. Display error if user enters invalid row/column numbers

➤ Answer:

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
{
    int row=5,col=5;
    char seat[row][col];
    int ch;
    for (int i = 0; i < row; i++)
    {
        for (int j = 0; j < col; j++)
        {
            seat[i][j] = 'O';
        }
    }
    do{
        cout << "Seating Arrangement:\n";
        for (int i = 0; i < row; i++)
        {
            for (int j = 0; j < col; j++)
            {
                cout << " " << seat[i][j] << " ";
            }
            cout << endl;
        }
    }
```

```
int rows,cols;
cout<<" Enter the row for booking (1-5)"<<endl;
cin>>rows;
cout<<"Enter the colon for booking (1-5)"<<endl;
cin>>cols;
```

```
if (rows >= 1 && rows <= 5 && cols >= 1 && cols <= 5)
{
    if (seat[rows-1][cols-1] == 'O')
```

```

    {
        seat[rows-1][cols-1] = 'X';
        cout << "Seat booked successfully!";
    }
    else
    {
        cout << "Sorry, pre occupies!";

    }
}

else
{
    cout << "This seat doesnt exist";
}

cout << "Updated Seating Arrangement:\n";
    for (int i = 0; i < row; i++)
    {
        for (int j = 0; j < col; j++)
        {
            cout << " " << seat[i][j] << " ";
        }
        cout << endl;
    }

    cout<<"Press 1 to book another seat?"<<endl;
    cin>>ch;
}while (ch==1);
return 0;
}

```

❖ Output

```
"C:\Users\beena\OneDrive\Dr  × + v
Seating Arrangement:
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
Enter the row for booking (1-5)
4
Enter the colon for booking (1-5)
2
Seat booked successfully!Updated Seating Arrangement:
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 X 0 0 0
0 0 0 0 0
Press 1 to book another seat?
1
Seating Arrangement:
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 X 0 0 0
0 0 0 0 0
Enter the row for booking (1-5)
3
Enter the colon for booking (1-5)
1
Seat booked successfully!Updated Seating Arrangement:
0 0 0 0 0
0 0 0 0 0
X 0 0 0 0
0 X 0 0 0
```

```
Seating Arrangement:
0 0 0 0 0
0 0 0 0 0
X 0 0 0 0
0 X 0 0 0
0 0 0 0 0
Enter the row for booking (1-5)
3
Enter the colon for booking (1-5)
1
Sorry, pre occupies!Updated Seating Arrangement:
0 0 0 0 0
0 0 0 0 0
X 0 0 0 0
0 X 0 0 0
0 0 0 0 0
Press 1 to book another seat?
|
```

